

STAT 20000 Chapter 21

The Accuracy of Percentages

In Ch 20, we learned how to set an appropriate box model,

$$\boxed{\text{some } 0\text{'s} \quad \text{some } 1\text{'s}} \quad |$$

when interested in percentage of subjects in a population with some favored property, a SRS drawn from the population can be analyzed with a box model.

The **sample percentage** of the SRS has the properties:

- ▶ $EV = \text{population percentage}$

$$\text{▶ } SE \approx \frac{\sqrt{\left(\frac{\text{fraction of } 1\text{'s in box}}{\text{fraction of } 0\text{'s in box}}\right) \left(\frac{\text{fraction of } 0\text{'s in box}}{\text{fraction of } 1\text{'s in box}}\right)}}{\sqrt{\text{sample size}}} \times 100\%,$$

when the sample size is small relative to the population size

- ▶ The normal curve can be used to figure chances about a sample percentage when the sample size is reasonably large.

Using the Sample to Infer About the Population

In reality, the population is not fully observed, and hence the **population percentage** is unknown.

unknown number of 0's unknown number of 1's

One can use the **sample percentage** to estimate the **population percentage**.

How accurate is this estimate?

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The *estimated* SE for the sample percentage is then

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The methods of replacing an unknown of the box with its counterpart of the sample are called “**bootstrap**”.

Example

In a certain city with 100,000 persons aged 18 to 24, a SRS of 500 such persons was drawn, of whom 194 turned out to be currently enrolled in college.

The percentage of all persons age 18 to 24 in that city who were currently enrolled in college is estimated as _____%, give or take _____% or so.

Example

In a certain city with 100,000 persons aged 18 to 24, a SRS of 500 such persons was drawn, of whom 194 turned out to be currently enrolled in college.

The percentage of all persons age 18 to 24 in that city who were currently enrolled in college is estimated as 38.8%, give or take _____% or so.

$$\blacktriangleright \text{estimate} = \text{sample percentage} = \frac{194}{500} \times 100\% = 38.8\%$$

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- ▶ estimate = sample percentage = $\frac{194}{500} \times 100\% = 38.8\%$
- ▶ the SD of sample is

$$\sqrt{\left(\frac{\text{fraction of } \boxed{1}\text{'s in sample}}{\boxed{1}}\right)\left(\frac{\text{fraction of } \boxed{0}\text{'s in sample}}{\boxed{0}}\right)} = \sqrt{\frac{194}{500} \left(1 - \frac{194}{500}\right)} = 0.487$$

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The percentage of all persons age 18 to 24 in that city who were currently enrolled in college is estimated as 38.8%, give or take 2.2% or so.

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- ▶ the bootstrap estimate of the SE is

$$\frac{\text{SD of sample}}{\sqrt{\text{sample size}}} \times 100\% \approx \frac{0.487}{\sqrt{500}} \times 100\% \approx 2.2\%$$

Example (Cont'd)

- ▶ The appropriate box model is: the percentage of college enrollees in the sample is like the percentage of _____'s among _____ draws made at random without replacement from the box

$$\underbrace{\left| \begin{array}{cc} ?? & \boxed{1} \end{array} \right| 's \quad \begin{array}{cc} ?? & \boxed{0} \end{array} \right| 's}_{\text{Box Model}}.$$

People in college are classified as $\boxed{}$, the others as $\boxed{}$.

Example (Cont'd)

- ▶ The appropriate box model is: the percentage of college enrollees in the sample is like the percentage of 1's among 500 draws made at random without replacement from the box

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People in college are classified as 1, the others as 0.

Confidence Intervals

Suppose there is some unknown population:

unknown number of 1's	unknown number of 0's
-----------------------	-----------------------

One can construct an interval for the plausible values of the unknown **population percentage** by the following procedure:

1. Take a **simple random sample** with a sufficiently large sample size
2. Compute the percentage of 1's in the sample
3. Estimate the SE using the bootstrap method
4. Construct the interval

sample percentage ± 1 SE

How much can one believe this interval contains the population percentage?

With a sufficiently large sample, from the normal approximation, we know

$$P\left(\left|\left(\text{sample percentage}\right) - \left(\text{population percentage}\right)\right| \leq 1 \text{ SE}\right) \approx 68\%.$$

Based on the frequency interpretation of chance, if the procedures on the previous slide is repeated a huge number of times, in about 68% of the times, *the sample percentage will lie between*

“the population percentage $\pm 1 \text{ SE}$ ”

or, equivalently, *the population percentage will lie between*

“the sample percentage $\pm 1 \text{ SE}$ ”

So the interval “sample percentage $\pm 1 \text{ SE}$ ” constructed based on the procedure on the previous slide is called a **68% confidence interval (68% C.I.)**.

► “68%” is the **confidence level**

Confidence Intervals at Other Confidence Levels

The procedure on page Ch21 - can be used to construct confidence intervals at other confidence levels:

90%-C.I. : sample percentage $\pm$ 1.64 SE

95%-C.I. : sample percentage $\pm$ 1.96 SE

99%-C.I. : sample percentage $\pm$ 2.58 SE

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Observe the trade off between *confidence* and *precision*:

- ▶ The greater the confidence you want to place in your prediction, the wider — and hence less informative — you have to make your interval

Interpretation of Confidence Intervals

For the example on page Ch21 - , a 95%-C.I. for the population percentage is

$$\begin{aligned} &\text{sample percentage} \pm \text{SE's} \\ &= \text{ } \% \pm \text{ } \% \approx 34.4\% \text{ to } 43.2\%. \end{aligned}$$

One may assert, **with 95% confidence, that the population percentage is between 34.4% and 43.2%.**

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Note we don't say

- ▶ “there is a 95% **chance** that the population percentage is between 34.4% and 43.2%,” or
- ▶ “there is a 95% **chance** that the interval (34.4%, 43.2%) covers the population percentage”

Why ?

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Why ?

- ▶ The population percentage is some fixed, albeit unknown, number. Either it does lie between 35% and 43%, or it does not. There is no “chance” involved.

Interpretation of Confidence Intervals (Cont'd)}

What is the thing that has a 95% chance to happen?

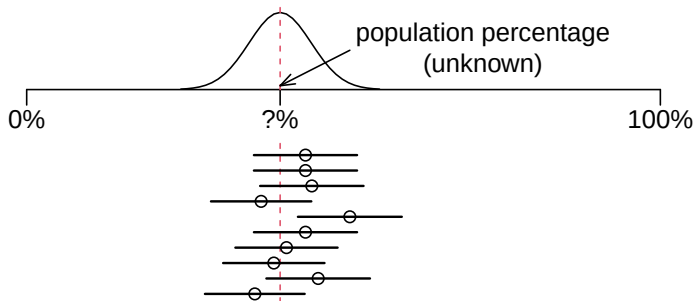
- ▶ It is the **procedure to construct the 95% interval**.
- ▶ 95% of the intervals constructed following the procedure on page Ch21 - will cover the true population percentage.
- ▶ After taking the sample and an interval is constructed, the constructed interval either covers the population percentage or it does not. We just don't know. Only God knows.

Demonstration

10 independent SRS's of size 100 were drawn from the box

unknown number of 1's unknown number of 0's

For each sample, a 90%-C.I. for the percentage of 1's in the box was constructed, with results shown below:



- ▶ The C.I.'s “move around” because they center at the sample percentage, which vary from sample to sample.

The Sample Size Issues

The **normal curve** can be used to figure chances about a sample percentage only when the **sample size is sufficiently large**.

The sample size required depends on the **box** (or the **population**), which is *unknown*.

For a 0-1 box, a **rule of thumb** is: the sample size n should be so large that

$$np \geq 10, \quad \text{and} \quad n(1 - p) \geq 10$$

where p = fraction of 1's in the **sample**.

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where p = fraction of 1's in the **sample**.

Example: How large the sample size need to be

- ▶ if sample percentage = 40%?
- ▶ if sample percentage = 10%?

True or False I

A box contains 10,000 marbles, of which some are red and the others blue. To estimate the percentage of red marbles in the box, 100 are drawn at random w/o replacement. Among the draws, 1 turns out to be red. The percentage of red marbles in the box is estimated as 1%, with an SE as

$$\frac{\sqrt{0.01 \times 0.99}}{\sqrt{100}} \approx 1\%.$$

True or false, and explain: An approximate 95% confidence interval for the percentage of red marbles in the box is $1\% \pm 2\% = (-1\%, 3\%)$.

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True or false, and explain: An approximate 95% confidence interval for the percentage of red marbles in the box is $1\% \pm 2\% = (-1\%, 3\%)$.

False. The sample size 100 is too small

$$(\text{sample size}) \times (\text{frac. of red in the sample}) = 100 \times \frac{1}{100} = 1 < 10.$$

There aren't enough draws to use the normal curve here.

True or False II

A simple random sample of 1,000 persons is taken from a large population and found 543 of them are Democrats. The sample percentage is $(543/1000) \times 100\% = 54.3\%$. The SE for the sample percentage is $\frac{\sqrt{0.543(1 - 0.543)}}{\sqrt{1000}} \times 100\% \approx 1.6\%$. True or false, and explain:

- ▶ $54.3\% \pm 3.2\%$ is a 95%-C.I. for the population percentage.
- ▶ $54.3\% \pm 3.2\%$ is a 95%-C.I. for the sample percentage.
- ▶ There is about a 68% chance for the percentage of Democrats in the population to be in the range $54.3\% \pm 1.6\%$.

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False. The sample percentage is known (but not of interest.)
We don't need a confidence interval for it.
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- ▶ There is about a 68% chance for the percentage of Democrats in the population to be in the range $54.3\% \pm 1.6\%$.
False. The population percentage either is in this particular interval, or is not in it — there is no “chance” involved.

True or False II (Cont'd)

True or false, and explain:

If another survey organization takes a SRS of 1,000 persons, there is about a 95% chance that the percentage of Democrats in their sample will be in the range $54.3\% \pm 3.2\%$.

Answer: _____.

The SE for a sample percentage measures the likely size of the difference between

the sample percentage and the population percentage,

not the difference between

two sample percentages obtained from two different samples.

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Gallup Polls

A Gallup poll pre-election survey based on a sample of 1,000 people estimates a 65% vote for the Democratic candidate in a certain election. True or false, and explain:

The likely size of the chance error in this estimate can be figured as follows —

$$\frac{\sqrt{0.65 \times 0.35}}{\sqrt{1,000}} \times 100\% = 1.5\% .$$

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$$\frac{\sqrt{0.65 \times 0.35}}{\sqrt{1,000}} \times 100\% = 1.5\% .$$

False. The Gallup poll isn't based on a **simple random sample**, so the formulas don't apply.

A psychologist administers a test of passivity to 100 students in his class and finds that 20 of them score over 50. He concludes that approximately 20% of all students would score over 50 on this test. He recognizes that this estimate may be off a bit, and he estimates the likely size of the error as follows:

$$\text{SE for percent} = \frac{\sqrt{0.2 \times 0.8}}{\sqrt{100}} \times 100\% = 4\%.$$

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- ▶ What population is he talking about?

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- ▶ Watch out for this guy!
- ▶ What population is he talking about?
- ▶ Are his students like a simple random sample from the population?

Until he can answer these questions, his calculation and conclusion are not very trustworthy.

Summary

- ▶ It is desired to estimate the percentage of people in a large population who have a given quality. This parameter is called the **population percentage**. It is unknown. A simple random sample is taken. The percentage of people in the sample who have the given quality is a statistic, called the **sample percentage**. The sample percentage is used to estimate the population percentage.
- ▶ The sample percentage will be off the population percentage, due to chance error. The SE for the sample percentage tells you the likely size of the amount off.
- ▶ When sampling from a 0-1 box whose composition is unknown, the SD of the box can be estimated by substituting the fractions of 0's and 1's in the sample for the unknown fractions in the box. This **bootstrap** estimate is good when the sample is reasonably large. The SD is needed to compute the SE.

Summary (Cont'd)

- ▶ A **confidence interval** for the population percentage is obtained by going the right number of standard errors either way from the sample percentage. The confidence level is read off the normal curve. This method should only be used with large samples.
- ▶ In the frequency theory of probability, parameters are not subject to chance variation. That is why confidence statements are made, instead of probability statements.
- ▶ The formulas for simple random samples may not apply to other kinds of samples. If the sample was not chosen by a probability method, watch out.